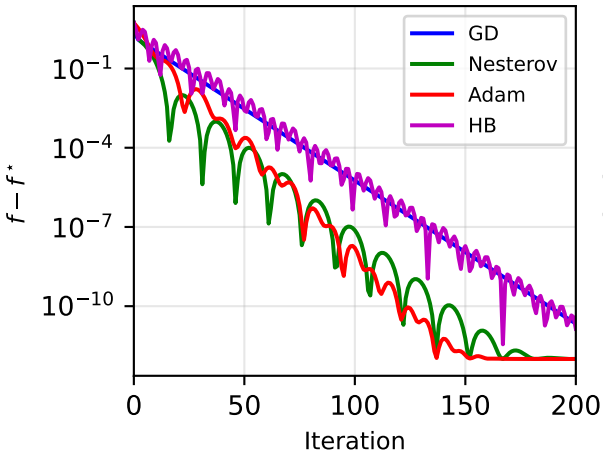
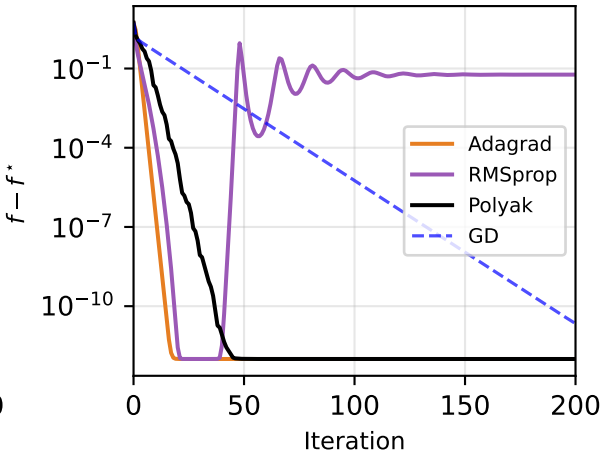


Q1-Q2 All-Method Performance Matrix: Q1 Adaptive vs Q2 Momentum/Stochastic (200 iterations)

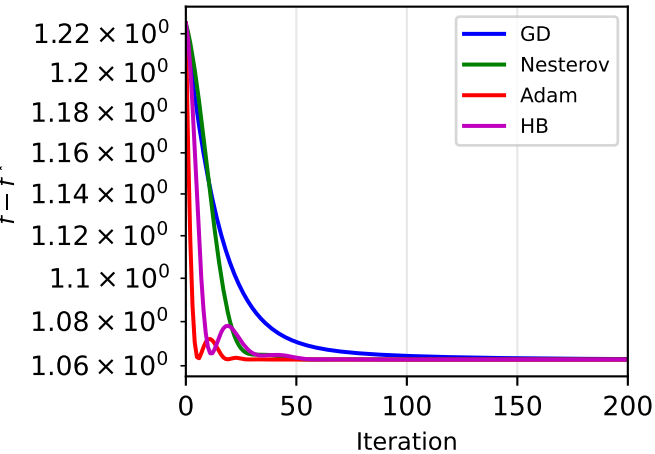
Bench. A: Quadratic



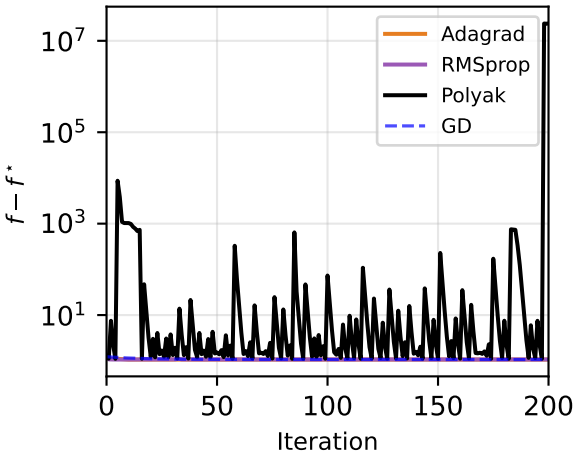
Bench. A: Adaptive Methods



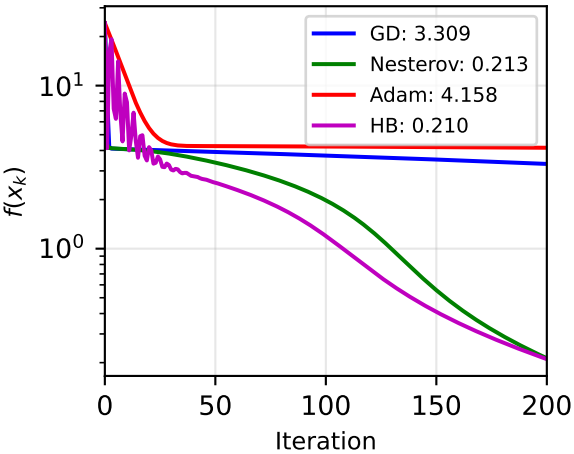
Bench. B: Toy NN



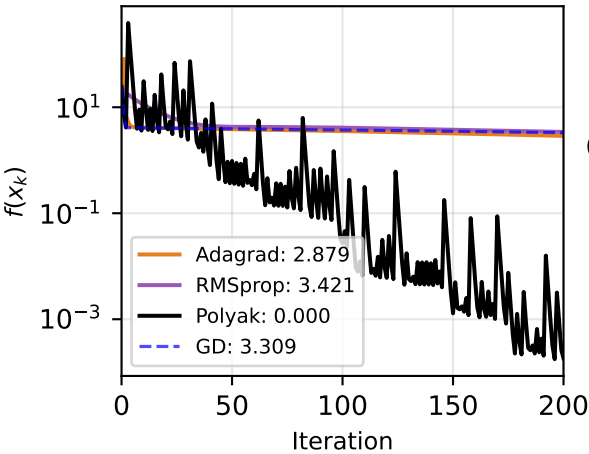
Bench. B: Adaptive Methods



Bench. C: Rosenbrock (Q2)



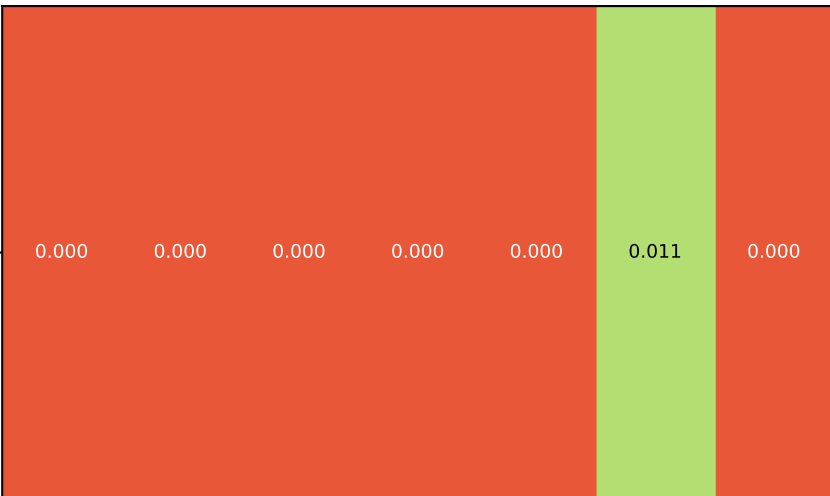
Bench. C: Rosenbrock (Q1)



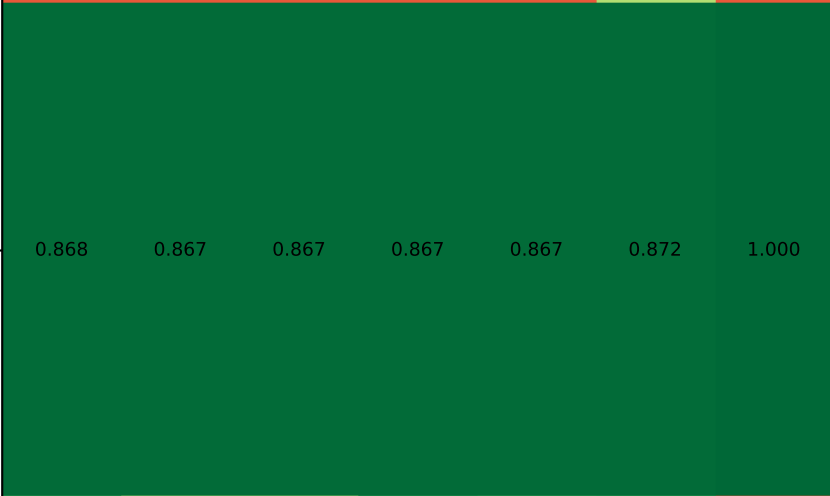
Normalised Suboptimality

$\log_{10}((f_T - f^*)/(f_0 - f^*))$

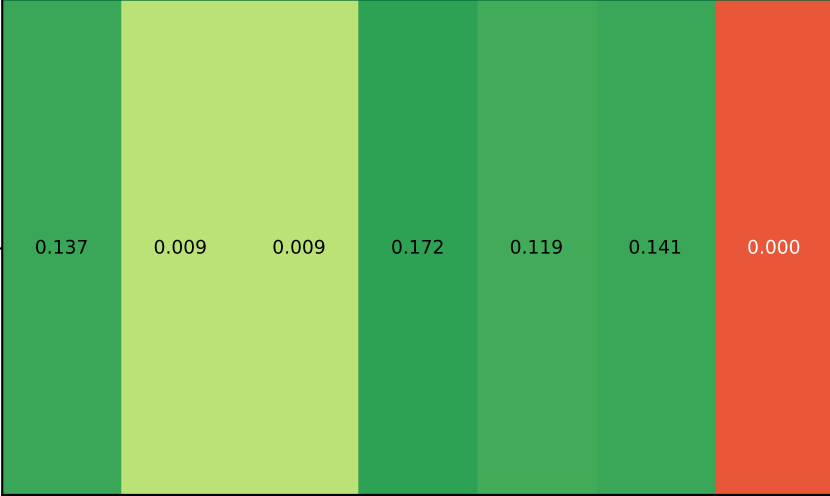
Bench. A
($\kappa \approx 1.09$)



Bench. B
($\kappa \leq 10$)



Bench. C
($\kappa \approx 2508$)



GD Heavy Ball Nesterov Adam Adagrad RMSprop Polyak

$\log_{10}$ normalised suboptimality

